First-line treatments for patients with advanced ALK gene rearrangements in NSCLC: a systematic review and network meta-analysis

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Supplementary Material

Supplementary Table 1. Literature search criteria

((((((((((NSCLC) OR (Carcinoma, Non Small Cell Lung)) OR (Carcinomas, Non-Small-Cell Lung)) OR (Lung Carcinoma, Non-Small-Cell)) OR (Lung Carcinomas, Non-Small-Cell)) OR (Non-Small-Cell Lung Carcinomas)) OR (Non-Small-Cell Lung Carcinoma)) OR (Non Small Cell Lung Carcinoma)) OR (Carcinoma, Non-Small Cell Lung)) OR (Non-Small Cell Lung Carcinoma)) OR (Non-Small Cell Lung Cancer)) OR (Nonsmall Cell Lung Cancer)) AND ((((((((((((((((((((((((crizotinib) OR (PF-02341066)) OR (PF-2341066)) OR (PF 2341066)) OR (PF2341066)) OR (PF 02341066)) OR (PF02341066)) OR (Xalkori)) OR (alectinib)) OR (Alecensa)) OR (CH5424802)) OR (RO5424802)) OR (ceritinib)) OR (5-chloro-N2-(2-isopropoxy-5-methyl-4-(piperidin-4-yl)phenyl)-N4-(2-(isopropylsulfonyl)phenyl)pyrimidine-2,4-diamine)) OR (Zykadia)) OR (LDK378)) OR (brigatinib)) OR (Alunbrig)) OR (AP26113)) OR (lorlatinib)) OR (PF-06463922)) OR (PF06463922)) OR (Lorbrena)) OR (7-amino-12-fluoro-2,10,16-trimethyl-15-oxo-10,15,16,17-tetrahydro-2H-8,4-(metheno)pyrazolo(4,3-h)(2,5,11) benzoxadiazacyclotetradecine-3-carbonitrile)) OR (ensartinib)) OR (6-amino-5-((1R)-1-(2,6-dichloro-3-fluorophenyl)ethoxy)-N-(4-(4-methylpiperazine-1-carbonyl)phenyl)pyridazine-3-carboxamide)) OR (X-396))

Supplementary Table 2. Checklist of the PRISMA extension for network meta-analysis

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Line 2
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Tab S2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Line 46-58
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Line 68-73
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Line 88-93
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Line 77-85
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Tab S1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Line 99-108
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Line 99-108
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Line 110-113
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any	Line 102-

Section and Topic	Item #	Checklist item	Location where item is reported
		assumptions made about any missing or unclear information.	108
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Line 95-97
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Line 110-113
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Fig. 1
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Line 101-108
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Tab. S3
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Line 131-141
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Line 142-146
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Line 142-146
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Line 146
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Line 197-199
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Line 150-153

Section and Topic	Item #	Checklist item	Location where item is reported
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Tab. S3
Study characteristics	17	Cite each included study and present its characteristics.	Tab. S3
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Fig. S1
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Fig. S1
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Fig. S4
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Tab. 1
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Line 192-197
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Fig.S6
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Line 196-197
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Line 197-201
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Line 210-215
	23b	Discuss any limitations of the evidence included in the review.	Line 282-298

Section and Topic	Item #	Checklist item	Location where item is reported
	23c	Discuss any limitations of the review processes used.	Line 282-298
	23d	Discuss implications of the results for practice, policy, and future research.	Line 216-281
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Line 85-86
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Line 85-86
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Line 85-86
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Line 319, Funding
Competing interests	26	Declare any competing interests of review authors.	Line 318, Competing interests
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Line 324, Data Sharing Statement

PRISMA = Preferred Reporting Items for Systematic Reviews and Meta-Analysis.

Supplementary Table 3. Baseline characteristics of the studies included in the network meta-analysis

Study (phase)	Intervention arm	Control arm	mPFS	mOS
PROFILE1014 (III)	Crizotinib,250 mg,BID	PbCT	10.9 vs. 7.0	NR vs. 47.5
ALEX (III)	Alectinib,600 mg,BID	Crizotinib,250 mg,BID	34.8 vs. 10.9	N/A vs. 57.4
J-ALEX (III)	Alectinib,300 mg,BID	Crizotinib,250 mg,BID	34.1 vs. 10.2	N/A vs. 43.7
ASCEND-4 (III)	Ceritinib,750 mg,QD	PbCT	16.6 vs. 8.1	N/A vs. 26.2
ALTA-1L (III)	Brigatinib,180 mg,QD	Crizotinib,250 mg,BID	24 vs. 11	N/A vs. N/A
PROFILE 1029 (III)	Crizotinib,250 mg,BID	PbCT	11.1 vs, 6.8	28.5 vs 27.7
ALESIA (III)	Alectinib,600 mg,BID	Crizotinib,250 mg,BID	N/A vs. 10.7	N/A vs. N/A
CROWN (III)	Lorlatinib,100 mg,QD	Crizotinib,250 mg,BID	N/A vs. 9.3	N/A vs. N/A
eXalt3 (III)	Ensartinib,225 mg，QD	Crizotinib,250 mg,BID	N/A vs. 12.7	N/A vs. N/A

N/A= not applicable; PbCT = pemetrexed-based chemotherapy;

Supplementary Table 4. Comparisons of the fit of consistency and inconsistency models using the deviance information criterion

	Progression-free survival				Overall survival	ORR	Grade ⩾3 AEs
Model	Overall	Baseline BM	No baseline BM	Asian	Overall	Overall	Overall
Consistency	**-16.83**	0.9853	**-11.78**	**-11.33**	**-6.67**	**124.4**	**116.1**
Inconsistency	-14.93	**-1.001**	-10.01	-9.39	-5.55	124.4	116.1

The deviance information criteria (DIC) is a Bayesian model evaluation criterion that measures model fit adjusted with complexity of the model; smaller DIC values correspond to more preferable models. (Reference: Spiegelhalter, D.J., Best, N.G., Carlin, B.P., V an der Linde, A. Bayesian measures of model complexity and fit. Journal of the Royal Statistical Society Series B (Statistical Methodology) 2002; 64(4):583-639).

Supplementary Table 5. Bayesian ranking results of the network meta-analysis

Treatment	Rank of possibility (%)						
	1	2	3	4	5	6	7
Progression-free survival							
PbCT	0	0	0	0	0	0	100
Crizotinib	0	0	0	0	92	8	0
Alectinib	6	58	30	6	0	0	0
Ceritinib	0	0	0	0	8	92	0
Brigatinib	1	9	30	59	0	0	0
Lorlatinib	**89**	8	2	1	0	0	0
Ensartinib	5	25	37	34	0	0	0
Overall survival							
PbCT	0	0	0	1	5	22	71
Crizotinib	0	1	6	25	41	25	2
Alectinib	**57**	30	10	3	1	0	0
Ceritinib	5	13	21	20	16	20	5
Brigatinib	5	13	22	21	16	15	9

Lorlatinib	27	29	19	11	6	5	3
Ensartinib	7	15	22	19	14	13	10
Objective response rate							
Alectinib	22	33	25	13	5	1	0
Brigatinib	15	18	20	19	15	10	2
Ceritinib	12	13	15	19	17	22	2
Crizotinib	0	1	5	17	37	40	0
Ensartinib	10	13	19	21	19	16	2
Lorlatinib	**41**	22	15	10	6	4	1
PbCT	0	0	0	0	1	5	93
Grade ≥3 adverse events							
Alectinib	0	0	1	3	10	86	-
Brigatinib	10	33	25	17	11	4	-
Ceritinib	**82**	12	4	1	1	1	-
Crizotinib	0	1	9	34	54	2	-
Lorlatinib	6	20	24	25	18	7	-
PbCT	2	35	37	20	6	1	-
Progression-free survival of patients with baseline brain metastases							
PbCT	0	2	3	7	30	57	-
Crizotinib	0	1	6	56	28	9	-
Alectinib	32	37	26	3	1	1	-
Ceritinib	4	5	8	22	34	27	-
Brigatinib	26	26	34	6	4	3	-
Lorlatinib	**38**	29	22	5	3	3	-
Progression-free survival of patients without baseline brain metastases							
PbCT	0	0	0	0	0	100	-
Crizotinib	0	0	1	70	29	0	-
Alectinib	19	77	3	0	0	0	-

Ceritinib	0	0	3	26	71	0	-
Brigatinib	0	4	91	4	1	0	-
Lorlatinib	**80**	19	1	0	0	0	-
Progression-free survival of Asian patients							
PbCT	0	0	0	0	5	95	-
Crizotinib	0	0	0	94	5	0	-
Alectinib	35	47	18	0	0	0	-
Ceritinib	0	0	0	5	90	5	-
Brigatinib	**47**	29	23	0	0	0	-
Lorlatinib	17	24	59	0	0	0	-

The number in each cell represents the posterior probability of the row-defining treatment being ranked at the column defining position. The numbers with biggest probability of ranking first are in bold and underscored. PbCT = pemetrexed-based chemotherapy;

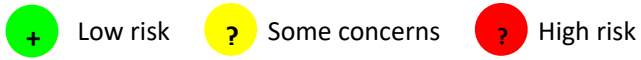
Supplementary Table 6. Result of sensitivity analysis

Treatment	Rank of possibility (%)						
	1	2	3	4	5	6	7
First sensitivity analysis of pooled PFS after excluded the J-ALEX study							
PbCT	0	0	0	0	0	0	100
Crizotinib	0	0	0	0	92	8	0
Alectinib	5	51	33	11	0	0	0
Ceritinib	0	0	0	0	8	92	0
Brigatinib	1	12	31	56	0	0	0
Lorlatinib	**89**	8	2	0	0	0	0
Ensartinib	5	29	34	32	0	0	0
Second sensitivity analysis of PFS in patients with baseline brain metastases after excluded the J-ALEX study							
PbCT	1	3	6	10	31	50	-
Crizotinib	0	3	11	51	27	9	-
Alectinib	27	34	27	6	3	3	-
Ceritinib	5	6	9	20	31	29	-
Brigatinib	26	27	28	8	5	6	-
Lorlatinib	**40**	27	19	6	4	4	-
Third sensitivity analysis of PFS in patients without baseline brain metastases after excluded the J-ALEX study							
PbCT	0	0	0	0	0	100	-
Crizotinib	0	0	1	70	29	0	-
Alectinib	18	76	5	0	0	0	-
Ceritinib	0	0	3	26	71	0	-
Brigatinib	0	6	89	4	1	0	-
Lorlatinib	**81**	18	1	0	0	0	-
Fourth sensitivity analysis of PFS in Asian patients after excluded the J-ALEX study							
PbCT	0	0	0	0	5	95	-
Crizotinib	0	0	0	94	5	0	-

Alectinib	30	44	25	0	0	0	-
Ceritinib	0	0	0	5	90	5	-
Brigatinib	**51**	29	21	0	0	0	-
Lorlatinib	19	27	54	0	0	0	-

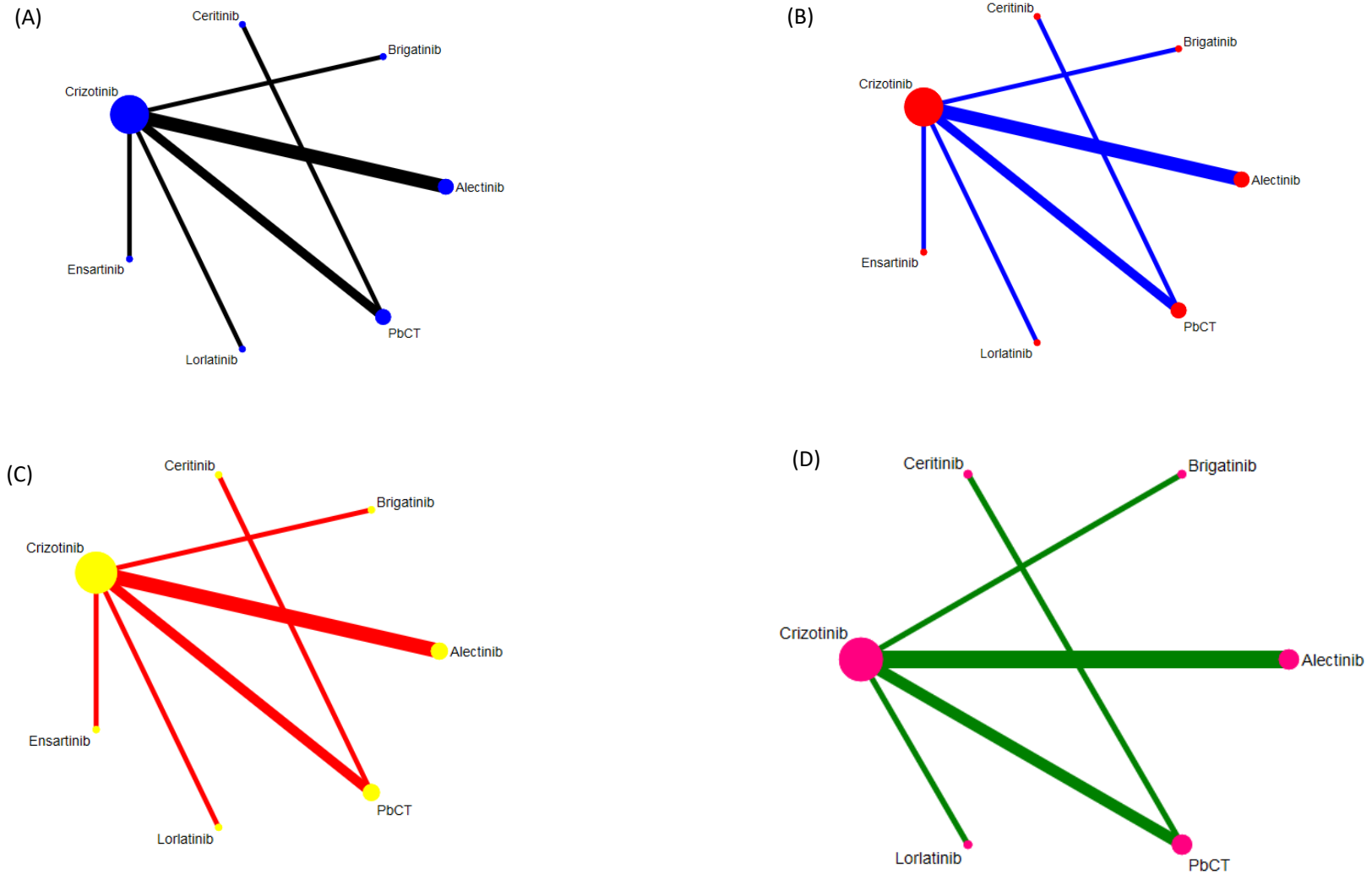
PbCT = pemetrexed-based chemotherapy;

Supplementary Figure 1. Summary of results from assessment of studies using the Cochrane risk of bias tool



	PROFILE1014	ALEX	J-ALEX	ASCEND-4	ALTA-1L	PROFILE 1029	ALESIA	CROWN	eXalt3
Randomization process	+	+	+	+	+	+	+	+	+
Deviations from intended interventions	?	+	+	+	?	+	?	+	?
Missing outcome data	+	+	+	+	+	?	+	+	+
Measurement of the outcome	?	+	+	+	+	+	+	+	+
Selection of the reported result	+	+	+	+	+	+	+	+	+
Overall	+	+	+	+	+	+	+	+	+

Supplementary Figure 2. Network diagrams of comparisons on different outcomes of treatments in different groups of patients. (A) PFS, (B) OS, (C) ORR, and (D) grade ≥3 adverse events



Supplementary Figure 3. Pooled hazard ratios of the PFS in patient subgroups. (A) Baseline brain metastases. (B) Baseline without brain metastases. (C) Asian patients

(A)

PbCT					
0.77 （0.34，2.00）	Crizotinib				
0.38 （0.12，1.12）	**0.50** **（0.21，0.95）**	Alectinib			
0.86 （0.25，2.85）	1.12 （0.24，4.46）	2.22 （0.46，11.65）	Ceritinib		
0.42 （0.09，2.05）	0.55 （0.16，1.70）	1.09 （0.29，4.85）	0.49 （0.08，3.24）	Brigatinib	
0.38 （0.10，1.77）	0.50 （0.16，1.65）	0.99 （0.29，4.43）	0.44 （0.07，3.17）	0.91 （0.18，5.05）	Lorlatinib

(B)

PbCT					
0.69 （0.62，0.77）	Crizotinib				
0.47 （0.4，0.55）	0.68 （0.6，0.77）	Alectinib			
0.73 （0.62，0.85）	1.06 （0.87，1.28）	1.56 （1.24，1.95）	Ceritinib		
0.57 （0.46，0.7）	0.83 （0.69，0.99）	1.22（0.98，1.51）	0.78（0.6，1.02）	Brigatinib	
0.42 （0.34，0.53）	0.61 （0.5，0.75）	0.9（0.72，1.14）	0.58（0.44，0.76）	0.74（0.57，0.96）	Lorlatinib

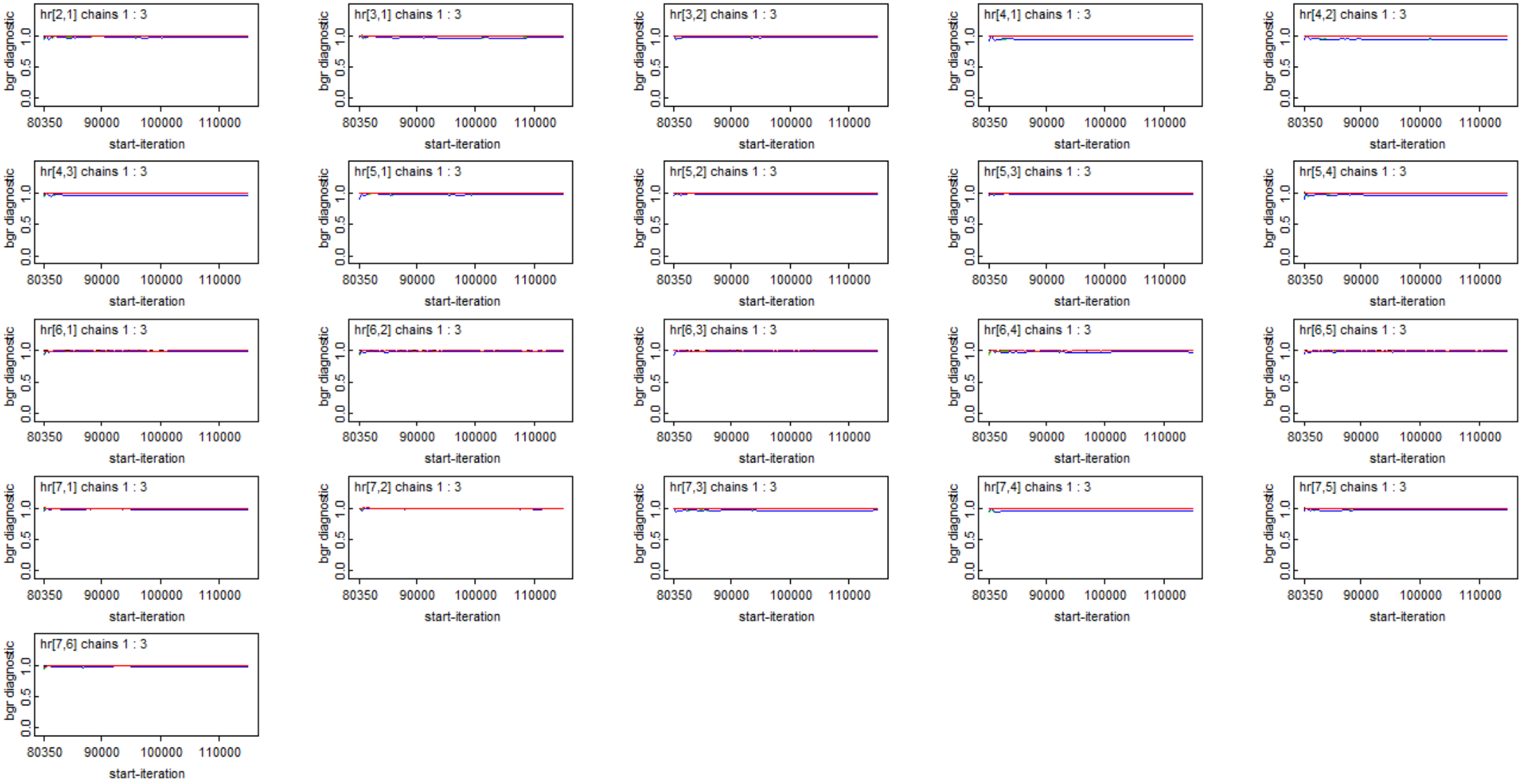
(C)

PbCT					
0.68 (0.61, 0.77)	Crizotinib				
0.45 (0.39, 0.53)	**0.67 (0.6, 0.74)**	Alectinib			
0.84 (0.67, 1.04)	1.22 (0.96, 1.57)	**1.84 (1.41, 2.41)**	Ceritinib		
0.45 (0.34, 0.58)	**0.66 (0.52, 0.83)**	0.99 (0.76, 1.28)	**0.54 (0.38, 0.75)**	Brigatinib	
0.49 (0.38, 0.64)	**0.72 (0.57, 0.91)**	1.08 (0.83, 1.4)	**0.59 (0.42, 0.83)**	1.09 (0.78, 1.53)	Lorlatinib

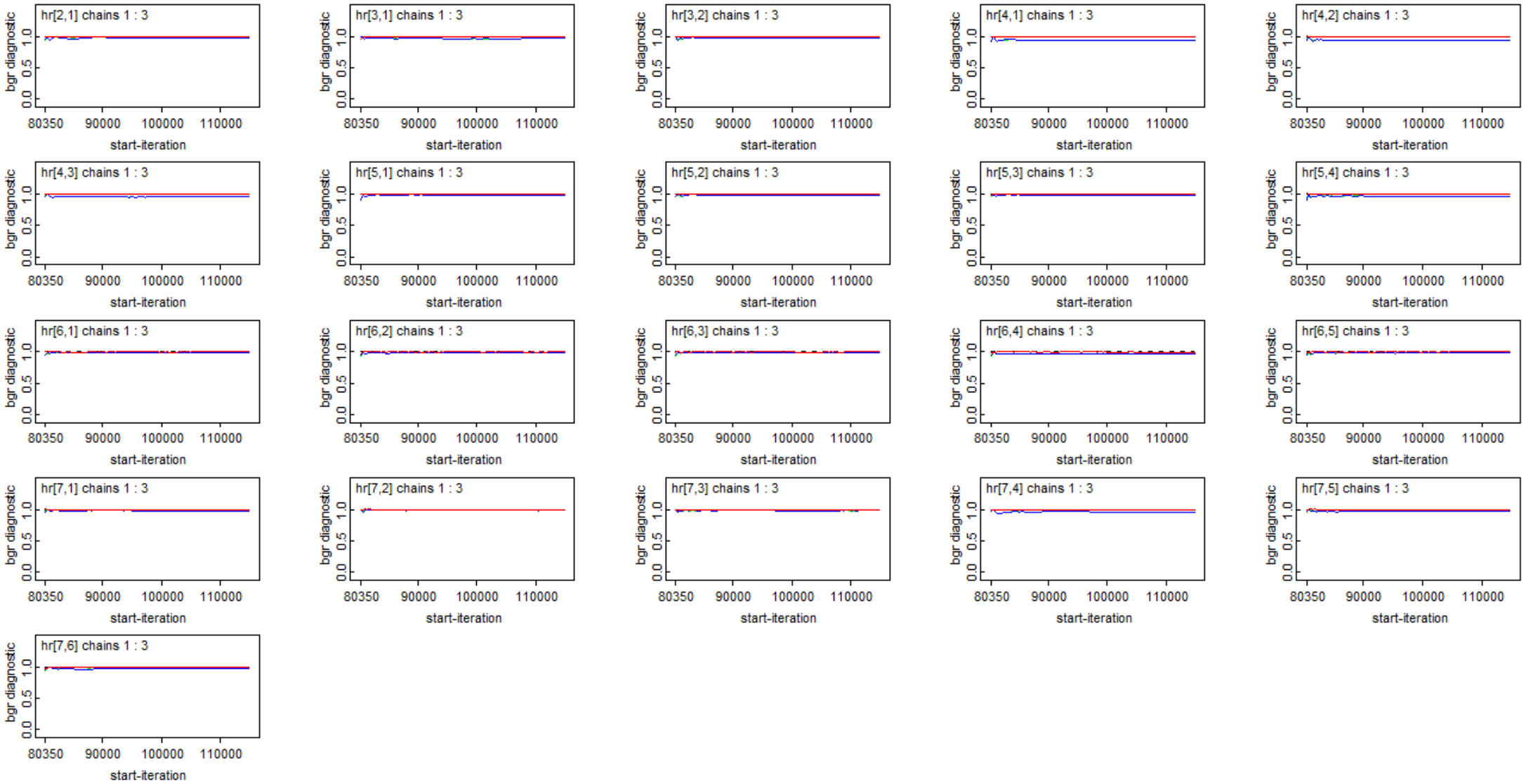
PbCT = pemetrexed-based chemotherapy;

Supplementary Figure 4. Convergence of the three chains established by inspection of the Brooks-Gelman-Rubin diagnostic.（A）PFS, （B）OS,（C）ORR, and（D）grade ≥3 adverse events.

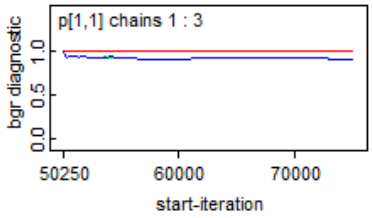
(A)

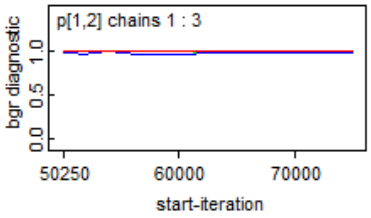


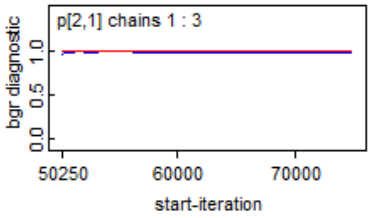
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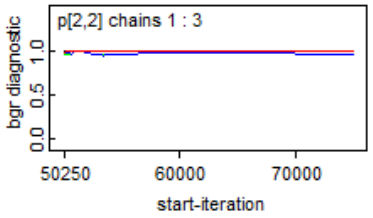

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bgr diagnostic
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0.5
1.0
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110000
start-iteration

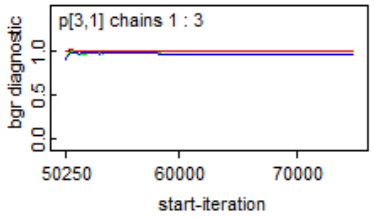
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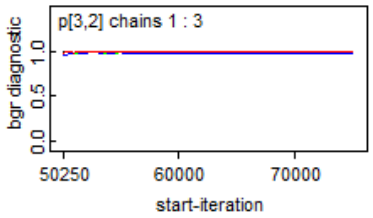

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start-iteration

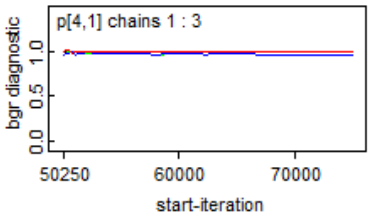

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start-iteration

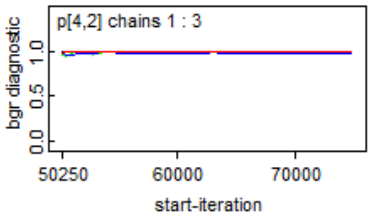

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50250 60000 70000
start-iteration

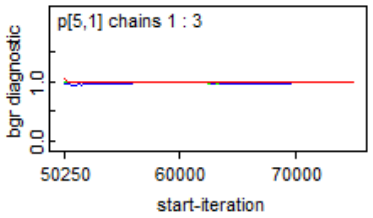

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50250 60000 70000
start-iteration

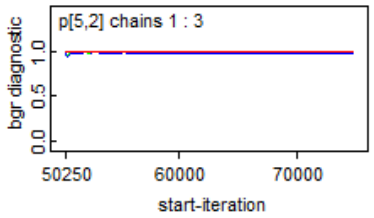

p[3,1] chains 1 : 3
bgr diagnostic
0.0 0.5 1.0
50250 60000 70000
start-iteration

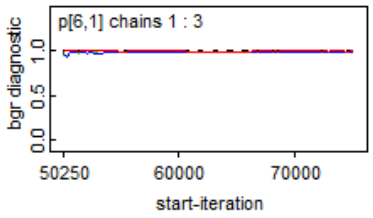

p[3,2] chains 1 : 3
bgr diagnostic
0.0 0.5 1.0
50250 60000 70000
start-iteration

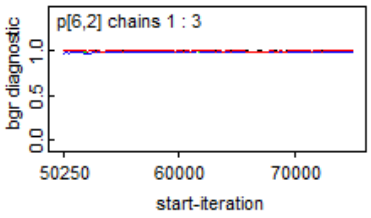

p[4,1] chains 1 : 3
bgr diagnostic
0.0 0.5 1.0
50250 60000 70000
start-iteration

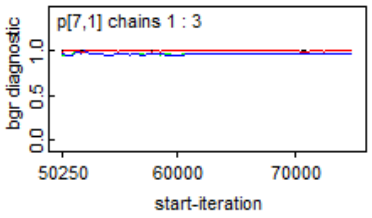

p[4,2] chains 1 : 3
bgr diagnostic
0.0 0.5 1.0
50250 60000 70000
start-iteration

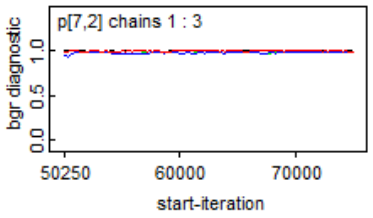

p[5,1] chains 1 : 3
bgr diagnostic
0.0 1.0
50250 60000 70000
start-iteration

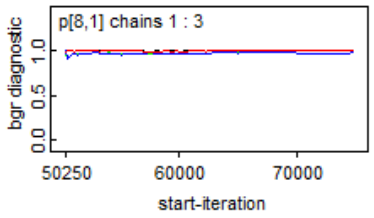

p[5,2] chains 1 : 3
bgr diagnostic
0.0 0.5 1.0
50250 60000 70000
start-iteration

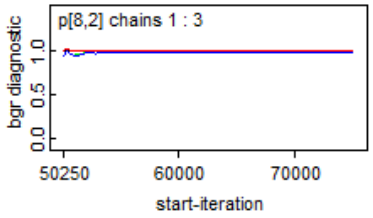

p[6,1] chains 1 : 3
bgr diagnostic
0.0 0.5 1.0
50250 60000 70000
start-iteration

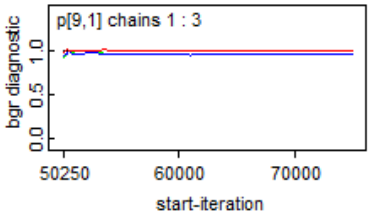

p[6,2] chains 1 : 3
bgr diagnostic
0.0 0.5 1.0
50250 60000 70000
start-iteration

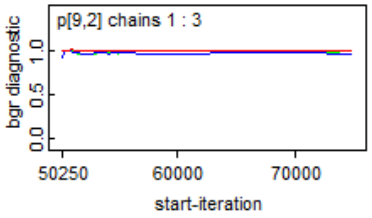

p[7,1] chains 1 : 3
bgr diagnostic
0.0 0.5 1.0
50250 60000 70000
start-iteration


p[7,2] chains 1 : 3
bgr diagnostic
0.0 0.5 1.0
50250 60000 70000
start-iteration

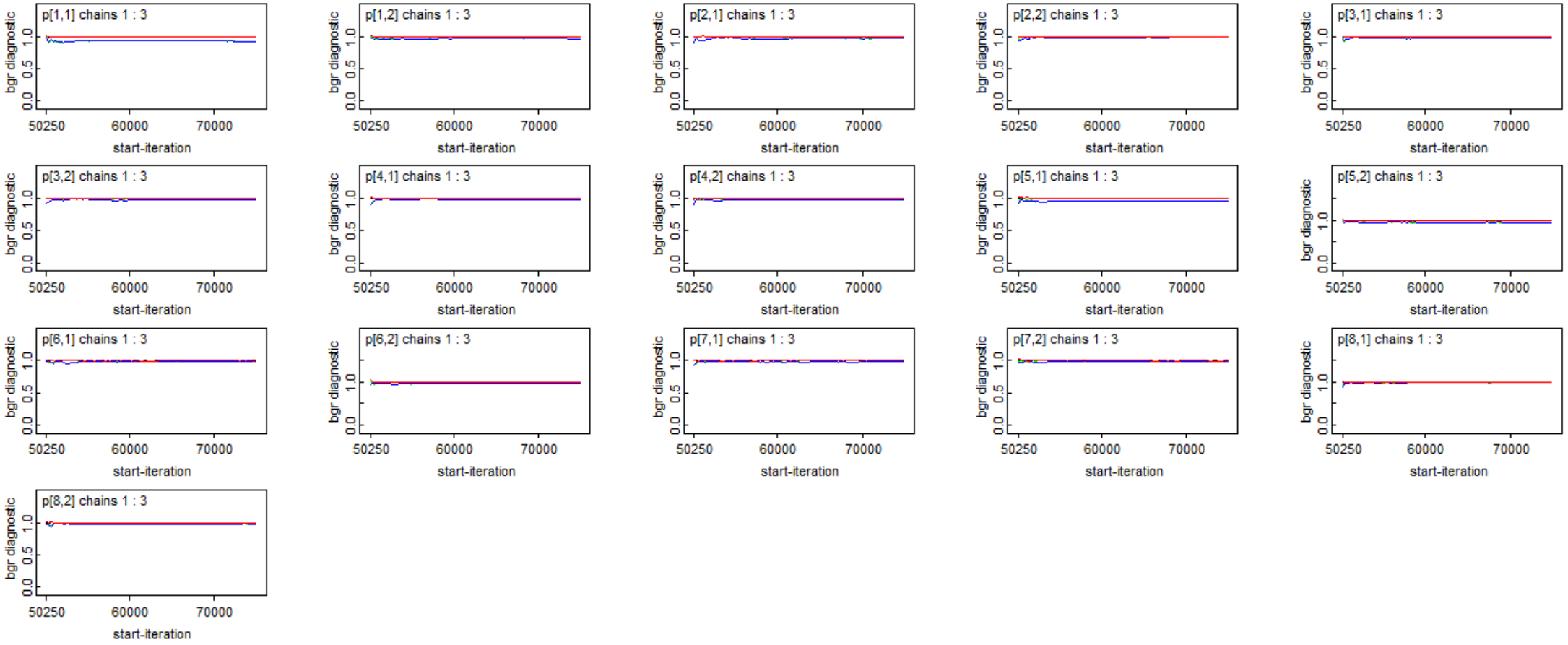

p[8,1] chains 1 : 3
bgr diagnostic
0.0 0.5 1.0
50250 60000 70000
start-iteration


p[8,2] chains 1 : 3
bgr diagnostic
0.0 0.5 1.0
50250 60000 70000
start-iteration


p[9,1] chains 1 : 3
bgr diagnostic
0.0 0.5 1.0
50250 60000 70000
start-iteration

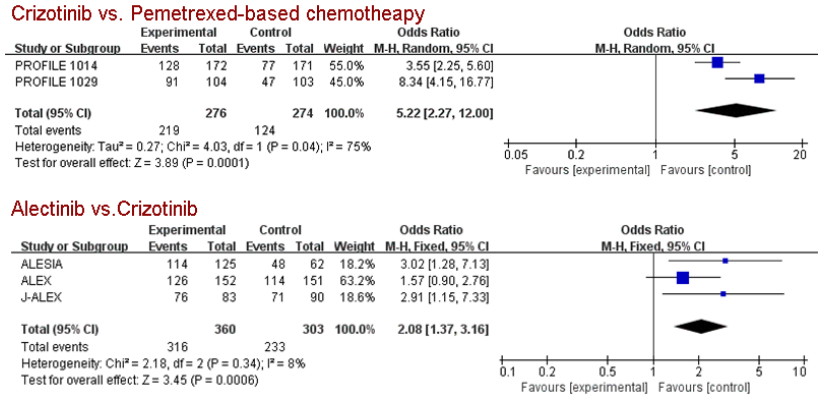

p[9,2] chains 1 : 3
bgr diagnostic
0.0 0.5 1.0
50250 60000 70000
start-iteration

(D)

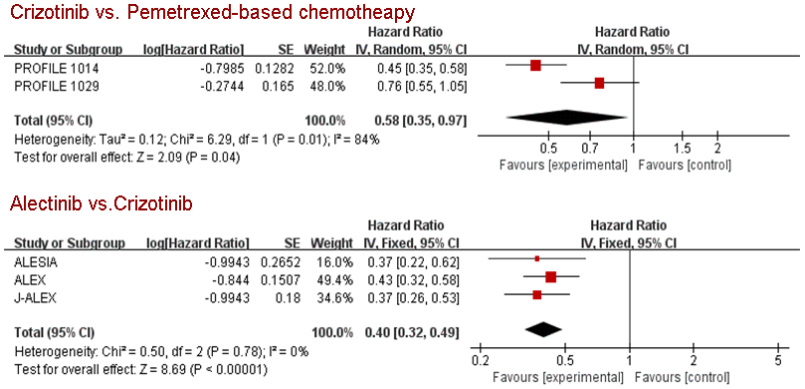

p[1,1] chains 1 : 3
p[1,2] chains 1 : 3
p[2,1] chains 1 : 3
p[2,2] chains 1 : 3
p[3,1] chains 1 : 3
p[3,2] chains 1 : 3
p[4,1] chains 1 : 3
p[4,2] chains 1 : 3
p[5,1] chains 1 : 3
p[5,2] chains 1 : 3
p[6,1] chains 1 : 3
p[6,2] chains 1 : 3
p[7,1] chains 1 : 3
p[7,2] chains 1 : 3
p[8,1] chains 1 : 3
p[8,2] chains 1 : 3
bgr diagnostic
0.0
0.5
1.0
50250
60000
70000
start-iteration

Supplementary Figure 5. Forest plots depicting results of head-to-head comparisons according to Bayesian pairwise and network meta-analyses.（A）ORR,（B）PFS,（C）OS, and（D）grade ≥ 3 adverse events

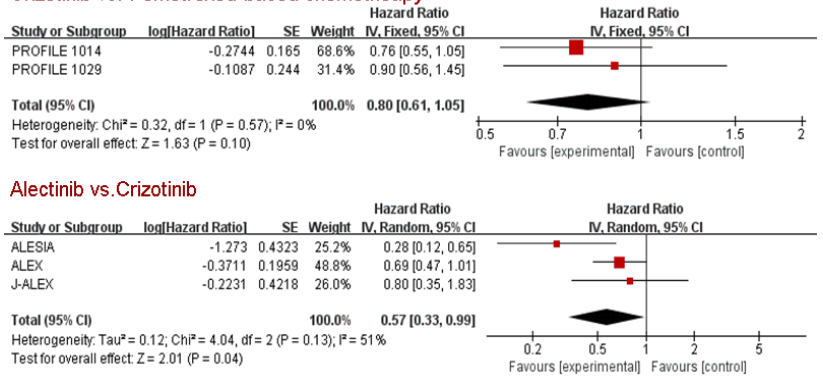
(A)



(B)



(C)



(D)

